

Derivatives:

$$f(x) = \ln\left(\sqrt{\frac{x^3+1}{x^3-1}}\right)$$

$$\Rightarrow f(x) = \frac{1}{2} [\ln(x^3+1) - \ln(x^3-1)]$$

$$f'(x) = \frac{1}{2} \left(\frac{2x}{x^3+1} - \frac{2x}{x^3-1} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{-2x(x^3+1) + 2x(x^3-1)}{(x^3+1)(x^3-1)} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{2x^3+2x-2x^3+2x}{(x^3+1)(x^3-1)} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{-4x}{(x^3+1)(x^3-1)} \right)$$

$$f'(x) = \frac{-2x}{(x^3+1)(x^3-1)}$$

Day 5

